

1 Domination in Graphs and the Removal of a Matching

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4 Abstract

5 We consider how the domination number of an undirected graph changes
6 on the removal of a maximal matching. It is straightforward that there are
7 graphs where no matching removal increases the domination number, and
8 where some matching removal doubles the domination number. We show
9 that in a nontrivial tree there is always a matching removal that increases the
10 domination number; and if a graph has domination number at least 2 there
11 is always a maximal matching removal that does not double the domination
12 number. We show that these results are sharp and discuss related questions.

13 1 Introduction

14 If G is a graph, then a *dominating set* D of G is a subset of vertices such that
15 every vertex of G is either in D or adjacent to a vertex in D . The *domination*
16 *number* of G , denoted $\gamma(G)$, is the cardinality of a minimum dominating set
17 of G , and a set achieving this is a γ -*set*. Note that in general we consider simple
18 undirected connected graphs.

19 One fundamental area of study as regards the domination number of a graph
20 is how it changes under graph operations. In particular, both the removal of
21 vertices, and the removal or addition of edges, have been the focus of many
22 papers. For example, vertex-critical graphs are ones where the deletion of any
23 vertex decreases the domination number (see for example [2]) while critical graphs
24 are ones where the addition of any edge decreases the domination number (see
25 for example [5]). One specific parameter in this regard is the *bondage number*
26 of a graph, which is defined to be the minimum number of edges whose removal
27 increases the domination number. For example, it is known that the bondage
28 number of a tree is at most 2 [3].

29 We study an extension of edge removal by considering what happens when
30 one removes a matching from a graph, where a *matching* is a set of edges such

31 that each vertex is incident to at most one edge. This is of course a special case
 32 of more general questions such as: what is the smallest k such that there exists
 33 a subset of edges, inducing a subgraph with maximum degree k , whose removal
 34 increases γ by at least ℓ ? But even for trees, the question of how much or
 35 how little the least-impact or the most-impact matching changes the domination
 36 number seems interesting. For example, we show that: (a) for a tree there always
 37 exists a matching whose removal increases the domination number (indeed any
 38 maximum matching suffices); (b) there are trees of arbitrarily large domination
 39 number where no matching removal increases it by more than 1; (c) in any graph
 40 a matching removal can at most double the domination number; and (d) while
 41 there is no graph with domination number bigger than 1 where for every maximal
 42 matching its removal doubles the domination number, there are trees of arbitrarily
 43 large domination number r where the removal of any maximal matching increases
 44 the domination number to at least $2r - 1$.

45 **2 The Fundamentals**

46 The removal of a matching can both change and not change the domination
 47 number. There are graphs, such as the 5-cycle, where removing any matching
 48 does not change the domination number. Or, for example, if the 2-domination
 49 number of a graph (meaning the smallest size of a set D such that every vertex
 50 outside D has at least two neighbor in D) equals its domination number, then
 51 the removal of a matching cannot change the domination number. But there are
 52 graphs, such as the complete graph K_n for n even, where the removal of any
 53 maximal matching does increase the domination number.

54 There is a simple limit though on how much the removal of a matching can
 55 increase the domination number. Since each vertex in a dominating set loses at
 56 most one neighbor, one immediately has:

57 **Lemma 1** *Removing a matching from a graph can at most double the domination*
 58 *number of the graph.*

59 For example, Figure 1 shows a tree T with $\gamma(T) = 3$, where there are maximal

60 matchings $M_3 = \{c, d\}$, $M_4 = \{c, e\}$, $M_5 = \{a, b, d\}$, and $M_6 = \{a, b, e\}$ such that
 61 $\gamma(T - M_i) = i$ for $3 \leq i \leq 6$.

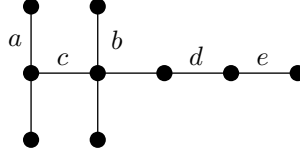


Figure 1: A tree with multiple maximal matchings

62 Now, in the case that one is removing a matching that has the most impact
 63 (that is, increases the domination number as much as possible), it is immediate
 64 that one may assume that the matching is maximal. On the other hand, in the
 65 case where one is removing a matching that has the least impact, it is necessary
 66 to add some condition on the matching, else the empty matching will be the
 67 answer. In this paper:

68 *The focus is on the case that the matching is maximal.*

69 Some of the results apply to all matchings, and so we state them as such. We
 70 also occasionally consider the case that the matching is required to be maximum,
 71 but this is a different problem. For example, Figure 2 shows a tree that has
 72 a unique maximum matching up to symmetry, and removing it increases the
 73 domination number from 3 to 5. On the other hand, removing the depicted
 74 maximal matching, consisting of two leaf-edges and the central edge, increases
 75 the domination number to 6. (We use *end-vertex* to mean a vertex of degree 1,
 76 and *leaf-edge* to mean an edge incident to such a vertex.)

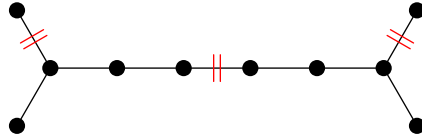


Figure 2: A tree where the matching with most impact is not maximum

77 There is a partial connection with the bondage number. If the bondage num-
 78 ber of a graph is 1, then it is immediate that the removal of any matching con-

79 taining that one edge increases the domination number. As noted above, a tree
80 has bondage number at most 2. But the two edges might need to be adjacent. In-
81 deed, if one has to remove independent edges, then there are trees T that require
82 the removal of $\gamma(T)$ edges to increase the domination number. One example is
83 the star where every edge has been subdivided once, as shown in Figure 3.

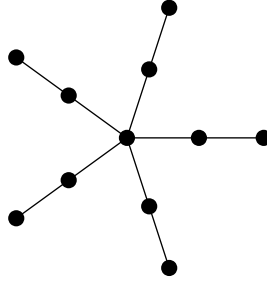


Figure 3: A subdivided star

84 It is well-known that the domination number of a graph without isolates is at
85 most half its order. One can ask a similar question for a matching-deleted graph:

86 **Theorem 1** *If G is a connected graph of order $n \geq 3$ and M a matching of G ,*
87 *then $\gamma(G - M) \leq 3n/4$.*

88 **PROOF.** Since the removal of an edge cannot decrease the domination number,
89 we may assume G is a tree. The proof is by induction on the order n . If G is a
90 star, then $\gamma(G - M) = 2$. So we may assume G is not a star. If G has diameter
91 3, then $\gamma(G) = 2$ and so $\gamma(G - M) \leq 4$. The bound follows if $n \geq 6$. But it can
92 be easily checked that the bound is true if n is 4 or 5. So we may assume that G
93 has diameter at least 4. Consider a longest path P in G with penultimate vertex
94 v .

95 *Case 1:* Vertex v has at least two end-vertex neighbors. Then let L be the
96 set consisting of v and its end-vertex neighbors. In $G - M$ it is still possible
97 to dominate L using at most two vertices. Let $G' = G - L$, and let M' be the
98 matching M restricted to $V(G')$. Since G has diameter at least 4, the graph G' has
99 at least three vertices. So by the induction hypothesis, $G' - M'$ can be dominated
100 with at most $3/4$ its order. It follows that $\gamma(G - M) \leq 2 + 3(n - |L|)/4 < 3n/4$.

101 *Case 2:* For all choices of P , vertex v has exactly one end-vertex neighbor.
 102 That is, it has degree 2 in G . Let w be the other neighbor of v , and x the other
 103 neighbor of w on P . If P has length 5 or more, then let J be the set of vertices
 104 separated from x by the bridge xw . Then let $G' = G - J$ and let M' be the
 105 matching M restricted to $V(G')$. By the induction hypothesis, $G' - M'$ can be
 106 dominated with at most $3/4$ its order. If P has length 4, then let $J = V(G)$ and
 107 vacuously the graph $G' = G - J$ can be dominated with at most $3/4$ its order.

108 Say J contains ℓ neighbors of w of degree 1 and t neighbors of w of degree 2.
 109 (Since we are in Case 2 it holds that $t \geq 1$.) Note that $|J| = 2t + \ell + 1$. If $\ell = 0$, in
 110 $G - M$ one can dominate J with at most $t + 1$ vertices; since $(t + 1) < 3(2t + 1)/4$
 111 it follows that then $\gamma(G - M) < 3|J|/4 + 3|G'|/4 = 3n/4$. If $\ell > 0$, in $G - M$ one
 112 can dominate J with at most $t + 2$ vertices; since $(t + 2) \leq 3(2t + 2)/4$ it follows
 113 that then $\gamma(G - M) \leq 3|J|/4 + 3|G'|/4 = 3n/4$. QED

114 The bound in Theorem 1 is best possible. Given a graph G , we define the k -
 115 *corona* of G as the graph obtained by, for each vertex, adding k new end-vertices
 116 adjacent to that vertex. If $k = 1$ we call it simply the corona. The value in
 117 Theorem 1 is achieved for the corona of a corona of a graph. Figure 4 shows an
 118 example where removing all leaf-edges increases the domination number to 15.

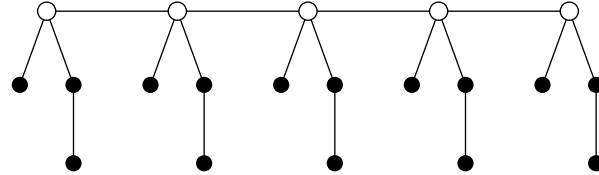


Figure 4: A corona of a corona

119 3 Matchings Whose Removal (Almost) Double the 120 Domination Number

121 In this section we provide some insight into the maximum change that removing
 122 a matching can cause. Consider for example a tree T where every vertex in the

123 (necessarily unique) γ -set has (at least) two end-vertex neighbors. Fink et al. [3]
 124 observed that for such a tree the bondage number is 1. We note that if M is a
 125 matching consisting of a leaf-edge incident with each vertex in the dominating
 126 set, it is immediate that $\gamma(T - M) = 2\gamma(T)$.

127 There is a bound on the domination number of graphs that have a matching
 128 whose removal doubles the domination number. We will need the following ter-
 129 minology: given a vertex set D and a vertex v of D , an *external private neighbor*
 130 of v with respect to D is defined to be a vertex outside D whose only neighbor
 131 in D is v .

132 **Theorem 2** *If a connected graph G with order $n \geq 3$ has a matching M such*
 133 *that $\gamma(G - M) = 2\gamma(G)$, then $\gamma(G) \leq n/3$.*

134 PROOF. Consider a γ -set D of G ; we can choose D to not contain an end-vertex.
 135 For each vertex $v \in D$, if v is incident with an edge of M then let v' be the end of
 136 that edge. Let the set X consist of D and all the v' . This set dominates $G - M$.
 137 By the hypothesis, this means $|X| = 2|D|$, so that all the vertices v' exist; that
 138 is, every vertex of D is incident with M . Further, the set X must be minimal
 139 dominating. So it follows that each v' has no neighbor in $D - v$; that is, in G the
 140 vertex v' is an external private neighbor of v with respect to D .

141 Let D' be the set of vertices of D that have only one external private neighbor
 142 in G . If D' is nonempty, consider any vertex $v \in D'$. By the choice of D , the
 143 vertex v is not an end-vertex and thus has a neighbor other than v' , say w .
 144 Since the set X is minimal dominating in $G - M$, and vertex v is in X only to
 145 dominate itself, it must be that w is not in D . Further, we claim that w has only
 146 one neighbor in D' . Since, if w also has neighbor $u \in D'$, one can take X and
 147 replace u, v by w and get a smaller dominating set, which is a contradiction.

148 For each vertex $v \in D'$, define the triple $R_v = \{v, v', w\}$. For each vertex
 149 $v \in D - D'$, define the triple R_v as v and two of its external private neighbors in
 150 G . By the above discussion these triples are disjoint. Thus the order of G is at
 151 least $3|D|$, whence the desired bound. QED

152 There are graphs and indeed trees that achieve the bound in Theorem 2. The

153 simplest example is the 2-corona of any graph; an example is shown in Figure 5.
 154 But another example is given in Figure 1.

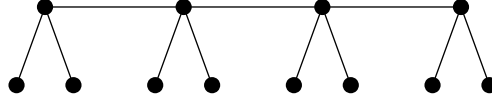


Figure 5: A 2-corona

155 It is also natural to look for graphs where the removal of any **maximum**
 156 matching doubles the domination number. Or even stronger, where the removal
 157 of any **maximal** matching doubles the domination number.

158 The graphs with domination number 1 are a special case. If a graph has
 159 domination number 1 and even order, then the removal of any maximal matching
 160 M increases the domination number to 2. For suppose there is still a dominating
 161 vertex v after the removal of M . Then v is not covered by M , and since the order
 162 is even, there is another uncovered vertex, necessarily a neighbor of v , which
 163 contradicts the maximality of M . If a graph has domination 1 and odd order, then
 164 both possibilities can occur. For example, the removal of any maximal/maximum
 165 matching from a star increases the domination number; but the removal of any
 166 maximal/maximum matching from a complete graph keeps it at 1.

167 So consider graphs with domination number at least 2. Here there are graphs
 168 where the removal of any maximum matching doubles the domination number.
 169 For example, define the octopus O_r by taking the star $K_{1,r}$ and subdividing every
 170 edge except one exactly three times. The result has $\gamma(O_r) = r$, achieved uniquely
 171 by taking the support vertices (where a support vertex is defined as one with an
 172 end-vertex neighbor). Further, the graph has a unique perfect matching, and its
 173 removal increases the domination number to $2r$. Figure 6 shows O_5 .

174 But the situation is (slightly) different if one considers all maximal matchings:

175 **Theorem 3** *If G is a connected graph with $\gamma(G) > 1$, then there exists a maximal*
 176 *matching M such that $\gamma(G - M) < 2\gamma(G)$.*

177 **PROOF.** Consider a γ -set D of G . If G is not a tree, choose a spanning tree T
 178 such that D dominates T . It then follows that there exist two vertices u and v

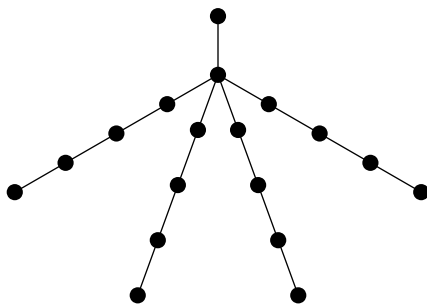


Figure 6: The octopus O_5

179 of D that are distance at most 3 apart in T , say joined by path P . Construct a
 180 maximal matching M by starting with the first edge of P and, if the length of P
 181 is 3, also the last edge of P . Extend to a maximal matching arbitrarily.

182 Create a dominating set D' of $T - M$ as follows. Start with $D' = D$. For
 183 every vertex of D other than u or v , they lose at most one neighbor when M
 184 is removed; if they do lose a neighbor, add that neighbor to D' . At this point
 185 $|D'| \leq 2|D| - 2$. The only possible vertices not dominated by D' are the neighbors
 186 of u and v in M ; call these u' and v' respectively. If u and v are adjacent in T ,
 187 then we constructed M to include the edge uv , and D' to include both u and v ,
 188 and so there is no undominated vertex. If u and v are at distance 2 in T , then
 189 u' is dominated by v , and so only v' is not dominated by D' , and we can add it
 190 to D' . If u and v are at distance 3, then u' and v' are the undominated vertices.
 191 But they are adjacent, and so can be dominated by adding one of them to D' .
 192 In all cases it follows for the final D' that $|D'| < 2|D|$. QED

193 The bound in Theorem 3 is best possible. That is, for all $r \geq 2$ there is a
 194 connected graph with $\gamma(G) = r$ and for every maximal matching M of G it holds
 195 that $\gamma(G - M) \geq 2r - 1$. For example, consider the octopus O_r defined above and
 196 a maximal matching M of O_r . By the maximality of M , it must contain an edge
 197 incident with every support vertex. It follows that a dominating set of $O_r - M$
 198 must contain at least two vertices from each subdivided edge, and one more vertex
 199 to dominate the end-vertex neighbor of the center. That is, $\gamma(O_r - M) \geq 2r - 1$.

200 4 Trees With Unique γ -Sets

201 We observed earlier that if a tree has a γ -set where every vertex is adjacent to
 202 at least two end-vertices, then the tree has a matching whose removal doubles
 203 the domination number. Such a tree is a special case of a tree with a unique
 204 γ -set. However, it is not the case that having a unique γ -set forces there to be a
 205 “doubling matching”. The smallest example is the path P_9 , where $\gamma = 3$ but the
 206 removal of a matching can only increase the domination number to 5.

207 It is immediate that if a graph has a unique γ -set, then the bondage number
 208 is 1 (first observed in [6]) and hence there is a matching whose removal increases
 209 the domination number. But one can say more in trees.

210 **Theorem 4** *If a tree T has a unique γ -set, then there is a matching M in T*
 211 *such that $\gamma(T - M) > \frac{3}{2}\gamma(T)$.*

212 This result is best possible as shown by the paths P_{3m} for m odd. Then
 213 $\gamma(P_{3m}) = m$, but removing a matching can only increase the domination number
 214 to $(3m + 1)/2$.

215 For the proof of Theorem 4, it is easier to work in a slightly more general
 216 setting. Gunther et al. [4] observed that in any graph if D is the unique γ -set
 217 (and there are no isolated vertices), then every vertex in D has (at least) two
 218 external private neighbors. Thus Theorem 4 follows from the following theorem.
 219 (Note that the statement of the theorem does not require D to be dominating.)

220 **Theorem 5** *If a tree T has a set D such that every vertex in D has at least two*
 221 *external private neighbors, then there is a matching M in T such that $\gamma(T - M) >$*
 222 *$\frac{3}{2}|D|$.*

223 **PROOF.** The proof is by induction on the order of T . Trivially we may assume
 224 D is not empty. Indeed:

225 **Claim 1** *We may assume that every support vertex is in D .*

226 *Proof.* The condition on D precludes an end-vertex x from being in D . So if
 227 such x is not dominated by D , just delete it from T and apply the induction
 228 hypothesis (and note that deleting an end-vertex cannot increase the domination
 229 number). $\llcorner$

230 Consider the base case. If each vertex in D has two neighbors that are end-
 231 vertices, then the matching M consisting of one leaf-edge incident with each
 232 vertex of D is such that $\gamma(T - M) \geq 2|D|$. It follows that we may assume
 233 the diameter of T is at least 5. If the diameter is exactly 5 and there is a
 234 vertex in D without two end-vertex neighbors, then let M consist of the central
 235 edge and one leaf-edge incident with each vertex of D , and again it follows that
 236 $\gamma(T - M) \geq 2|D|$. So we may assume the diameter is at least 6.

237 For the inductive step, most of the time we proceed as follows. We identify a
 238 subset S of the vertices such that $T' = T - S$ is a tree that satisfies the hypothesis
 239 with $D - S$. The inductive hypothesis yields a matching M' of T' , which we extend
 240 to a matching M of T by adding a set N of edges all of whose ends are in S .
 241 Further, S and M are such that the vertices of T' that have a neighbor in S can
 242 be dominated in $T - M$ by one vertex of $V(T')$. We say S and M with all the
 243 above properties are *standard*.

244 For any subset S and matching M , let ψ_S^M denote the minimum possible
 245 number of vertices of S in any dominating set of $T - M$.

246 **Claim 2** *Let S and M be standard.*

- 247 (a) *It holds that $\gamma(T - M) \geq \psi_S^M + \gamma(T' - M') - 1$.*
 248 (b) *If $\psi_S^M \geq \frac{3}{2}|D \cap S| + 1$, then the inductive step is valid.*
 249 (c) *Further, if no set that achieves ψ_S^M has a neighbor in $V(T')$, then for the*
 250 *inductive step it is sufficient that $\psi_S^M \geq \frac{3}{2}|D \cap S|$.*

251 *Proof.* (a) Consider a γ -set of $T - M$ and let X be the restriction of it to $V(T')$.
 252 Then $|X| \leq \gamma(T - M) - \psi_S^M$. Further, X dominates all of $T' - M$ except possibly
 253 those vertices of T' with neighbors in S , and these can be dominated by one
 254 vertex of $V(T')$ by the definition of standard. Thus $\gamma(T' - M') \leq |X| + 1$. It
 255 follows that $\gamma(T - M) - \psi_S^M \geq \gamma(T' - M') - 1$, which re-arranged gives one the
 256 desired inequality.

257 (b) We have that $\gamma(T - M) \geq \psi_S^M + \gamma(T' - M') - 1 > \psi_S^M + \frac{3}{2}(|D - S|) - 1 =$
 258 $\frac{3}{2}|D| + \psi_S^M - \frac{3}{2}|D \cap S| - 1$. The claim follows.

259 (c) The calculation is similar, except that $\gamma(T - M) \geq \psi_S^M + \gamma(T' - M')$. $\triangleleft$

260 It is common in trees to induct by focussing on the vertices at the end of
 261 a longest path. We generalize this slightly. Given a pair of adjacent vertices
 262 v and w , we define a (v, w) -*peripheral path* as a path starting with edge vw
 263 of longest possible length, and call its length the (v, w) -peripheral length. For
 264 a vertex v that is not an end-vertex, we define the *peripherality* of v , denoted
 265 $per(v)$, by considering the multiset of (v, w) -peripheral lengths for all neighbors
 266 w , and taking the **second-largest** length (which might equal the largest). We
 267 define $per(v) = 0$ if v is an end-vertex. For example, if $v_0v_1 \dots v_d$ is a longest
 268 path in T , then $per(v_i) = per(v_{n-i}) = i$ for $0 \leq i \leq d/2$. Further, we designate
 269 the neighbor w of v with the largest (v, w) -peripheral length its *free* neighbor.
 270 Note that, if the diameter of T is more than $2 per(v)$, then the free neighbor of v
 271 is uniquely determined; otherwise designate one arbitrarily, if necessary.

272 We continue the proof of the theorem. Since the diameter is at least 6 there is
 273 some vertex v_3 with $per(v_3) = 3$. Note that in all the figures in the proof of this
 274 theorem, solid vertices are vertices that are definitely in D and hollow vertices
 275 are vertices that are definitely not in D .

276 **Claim 3** *If $per(v_3) = 3$ with peripheral path $v_3v_2v_1v_0$, then we may assume both*
 277 *v_1 and v_2 have degree 2.*

278 *Proof.* By Claim 1, $v_1 \in D$.

279 (i) We first prove that v_1 has degree 2. So suppose that v_1 has at least two
 280 end-vertex neighbors. There are three cases.

281 *Case A: Assume v_2 is in D and v_2 has two end-vertex neighbors.* Then
 282 induct with S consisting of all vertices separated from v_2 by the bridge v_1v_2 ,
 283 and $N = \{v_0v_1\}$. (See Figure 7a for an example.) It is immediate that $\psi_S^M = 2$
 284 while $|D \cap S| = 1$. Since one may assume that v_2 is in the dominating set of
 285 $T - M$ (as it has an end-vertex neighbor), the bound holds by Claim 2b.

286 *Case B: Assume v_2 is in D but has only one end-vertex neighbor.* Then v_3
 287 must be the other external private neighbor of v_2 . Let v_4 be the free neighbor of

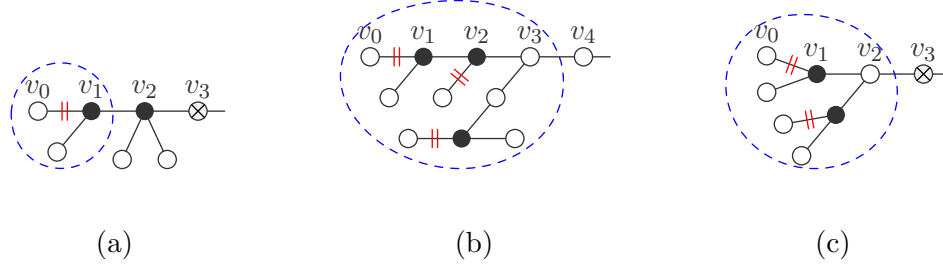


Figure 7: Three possible choices of S

288 v_3 . Then, by the peripherality of v_3 , all other neighbors of v_3 , if any, are within
 289 distance 2 of an end-vertex. Since v_3 has no other neighbor in D and is not in D
 290 itself, it follows that each of these neighbors must be distance exactly 2 from an
 291 end-vertex, have a neighbor in D and all other neighbors of that neighbor must
 292 be end-vertices. (See Figure 7b for an example.) Induct with S consisting of all
 293 vertices separated from v_4 by the bridge v_3v_4 , and N consisting of one leaf-edge
 294 incident with each vertex in $D \cap S$. Then every vertex of $D \cap S$ except v_2 has two
 295 end-vertex neighbors in T and hence at least one end-vertex neighbor in $T - M$;
 296 thus $\psi_S^M = 2|D \cap S| - 1$. However, any set achieving this does not dominate v_4 .
 297 So the bound holds by Claim 2b, since $|D \cap S| \geq 2$.

298 *Case C: Assume $v_2 \notin D$.* By the peripherality of v_3 , it must be that any
 299 neighbor of v_2 , other than v_3 , is in D and has at least two end-vertex neighbors.
 300 (See Figure 7c for an example.) Induct with S consisting of all vertices separated
 301 from v_3 by the bridge v_2v_3 , and N consisting of one leaf-edge incident with each
 302 vertex of $D \cap S$. We have $\psi_S^M = 2|D \cap S|$; but a set achieving this cannot dominate
 303 v_3 . So the bound holds by Claim 2b.

304 Hence we have shown that one may assume that v_1 has degree 2.

305 (ii) It follows that v_2 is the external private neighbor of v_1 . Thus v_2 cannot
 306 have another neighbor with peripherality at most 1, and so it has degree 2. $\triangleleft\triangleleft$

307 It follows that a vertex of peripherality 3 cannot be in D .

308 **Claim 4** *We may assume that every vertex v_3 of peripherality 3 has degree 2.*

309 *Proof.* Let $v_3v_2v_1v_0$ be a peripheral path, and let v_4 denote the free neighbor
 310 of v_3 . Suppose v_3 has a third neighbor w .

311 (i) Assume that w has peripherality 2. Then by Claim 3, w has degree 2 and
 312 its other neighbor has degree 2 too. Then induct with S consisting of the vertices
 313 separated from v_3 by the bridge v_2v_3 together with the vertices separated from
 314 v_3 by the bridge wv_3 , and N consisting of a leaf-edge incident with each vertex
 315 of $D \cap S$. (See Figure 8a.) It follows that $\psi_S^M = 4$ while $|D \cap S| = 2$, and so the
 316 bound holds by Claim 2a.

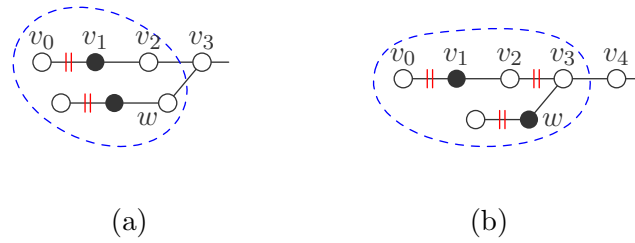


Figure 8: Two possible choices of S

317 (ii) Assume that w has peripherality 1 and two end-vertex neighbors. Then
 318 induct as in the previous case.

319 (iii) Assume v_3 has no neighbor of these two types. It follows that w has
 320 peripherality 1 and only one end-vertex neighbor, and thus that v_3 is an external
 321 private neighbor of w . In particular, v_3 has degree 3. (See Figure 8b.) Then
 322 induct with S consisting of the vertices separated from v_4 by the bridge v_3v_4 ,
 323 and N consisting of v_0v_1 , v_2v_3 , and the leaf-edge incident with w . It follows that
 324 $\psi_S^M = 4$ while $|D \cap S| = 2$, and so the bound holds by Claim 2b. $\llcorner$

325 If the tree has diameter 6 or 7, then it follows from the above claim that T
 326 is a path, that is, P_7 or P_8 . It is easy to observe that the largest D satisfying
 327 the hypothesis of the theorem has two vertices, and that removing a maximum
 328 matching yields a forest with domination number at least 4. So we may assume
 329 that T has diameter at least 8. In particular, there exists some vertex v_4 with
 330 $per(v_4) = 4$.

331 **Claim 5** *We may assume that every vertex v_4 of peripherality 4 is in D and has*

332 degree 2.

333 *Proof.* Let $v_4v_3v_2v_1v_0$ be a peripheral path.

334 (i) Suppose that $v_4 \notin D$. Then induct with $S = \{v_0, v_1, v_2, v_3\}$ and $N =$
 335 $\{v_0v_1, v_2v_3\}$. Then $\psi_S^M = 2$ and no set achieving this contains v_3 , while $|D \cap S| =$
 336 1. Thus the bound follows from Claim 2c. So we may assume v_4 is in D .

337 (ii) Suppose v_4 has degree more than 2. Let v_5 be its free neighbor. Let m
 338 be the length of the longest path that starts with v_4 and does not use v_3 or v_5 .
 339 Since $\text{per}(v_4) = 4$, it follows that $m \leq 4$. There are four cases.

340 *Case A:* $m = 4$. Say we have path $v_4w_3w_2w_1w_0$. Then $\text{per}(w_3) = 3$, and
 341 so by the above claim has degree 2. We use a non-standard inductive step. Let
 342 $S = \{v_0, v_1, v_2, w_0, w_1, w_2\}$. (See Figure 9a.) Then induct on $T' = T - S$ to
 343 produce matching M' . Then, since M' can contain only one edge incident with
 344 v_4 , we may assume without loss of generality that the edge v_3v_4 is not in M' .
 345 So we can define $N = \{v_0v_1, v_2v_3, w_0w_1\}$. It follows that $\psi_S^M = 4$, but only one
 346 vertex of T' can be dominated by S in $T - M$. Thus $\gamma(T - M) \geq \gamma(T' - M') + 3$,
 347 while $|D \cap S| = 2$, and so the bound follows.

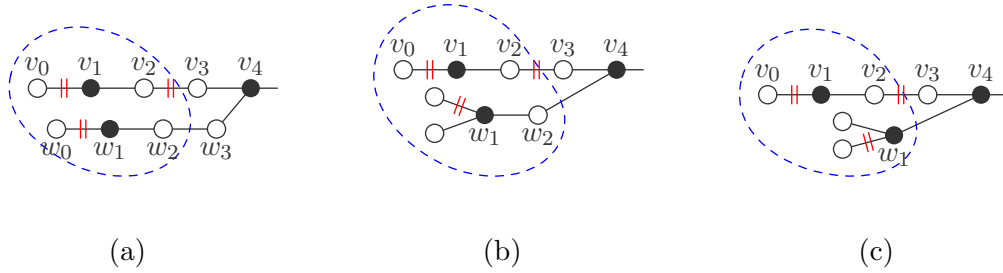


Figure 9: Three possible choices of S

348 *Case B:* $m = 3$. Say we have path $v_4w_2w_1w_0$. Then w_1 is a support vertex
 349 and so it is in D ; and thus w_1 has (at least) two end-vertex neighbors, while w_2
 350 is not an external private neighbor of v_4 . Indeed, $\text{per}(y) \leq 1$ for all neighbors
 351 y of w_2 other than v_4 . Again we use a non-standard inductive step. Let S
 352 consist of v_0, v_1, v_2 together with all vertices separated from v_4 by the bridge w_2v_4 .
 353 (See Figure 9b.) Then induct on $T' = T - S$ to produce matching M' . Now

354 form M^* from M' by deleting edge v_3v_4 if present. Then form M from M^*
355 by adding v_2v_3 and one leaf-edge incident with each vertex of $D \cap S$. Then
356 $\psi_S^M = 2|D \cap S|$. But since in $T - M$ vertex v_3 is an end-vertex adjacent to v_4 , we
357 may assume v_4 is in the dominating set of $T - M$; thus a set achieving ψ_S^M cannot
358 help with dominating $V(T')$. It follows that $\gamma(T - M) \geq \psi_S^M + \gamma(T' - M^*) \geq$
359 $\psi_S^M + \gamma(T' - M') - 1$, and the bound follows since $|D \cap S| \geq 2$.

360 *Case C: $m = 2$.* Say we have path $v_4w_1w_0$. Then w_1 is in D and has (at least)
361 two end-vertex neighbors. Again we use a non-standard inductive step. Let S
362 consist of v_0, v_1, v_2 together with all vertices separated from v_4 by the bridge w_1v_4 .
363 (See Figure 9c.) Then induct on $T' = T - S$ to produce matching M' . Now, form
364 M^* from M' by deleting edge v_3v_4 if present. Then form M from M^* by adding
365 edges v_0v_1, v_2v_3 , and w_0w_1 . Then $\psi_S^M = 4$ while $|D \cap S| = 2$; and by a similar
366 argument to the previous case, it follows that $\gamma(T - M) \geq \gamma(T' - M') + 3$, so
367 that the bound follows similarly.

368 *Case D: $m = 1$.* That is, all other neighbors of v_4 are end-vertices. Then
369 induct with S consisting of all vertices separated from v_5 by the bridge v_4v_5 , and
370 N consisting of a leaf-edge incident with each of v_1 and v_4 . Again $\psi_S^M = 4$ while
371 $|D \cap S| = 2$. Thus the claim holds. $\llcorner$

372 If the tree has diameter 8 or 9, then it follows from the above claim that T
373 is a path, that is, P_9 or P_{10} . It is easy to observe that the largest D satisfying
374 the hypothesis of the theorem has three vertices, and that removing a maximum
375 matching yields a forest with domination number at least 5. So we may assume
376 that T has diameter at least 10. In particular, there exists some vertex v_5 with
377 $\text{per}(v_5) = 5$.

378 Say v_5 has peripheral path $v_5v_4v_3v_2v_1v_0$ and free neighbor v_6 . Let y be a
379 neighbor of v_5 other than v_6 , if it exists. Then $\text{per}(y) \leq 4$. Since v_5 cannot have
380 a second neighbor in D , it follows that y is not in D . Hence by the above claim
381 $\text{per}(y) \neq 4$. Also y does not have an end-vertex neighbor. It follows that, for
382 each neighbor z of y apart from v_5 , $\text{per}(z) \leq 2$. If $\text{per}(z) = 2$ then $\text{per}(y) = 3$,
383 and so z is not in D . (See Figure 10.) If there is a z with $\text{per}(z) = 1$ that does
384 not have at least two end-vertex neighbors, then it is the only z with $\text{per}(z) = 1$.
385 Induct with S consisting of all vertices separated from v_6 by the bridge v_5v_6 , and

386 N consisting of v_0v_1 , v_2v_3 , v_4v_5 , and one leaf-edge incident with each vertex in
 387 $(D \cap S) - \{v_1, v_4\}$. Then $\psi_S^M = 2|D \cap S| - 1$ and any set achieving this does not
 388 contain v_5 . So the bound follows from Claim 2c, since $|D \cap S| \geq 3$.

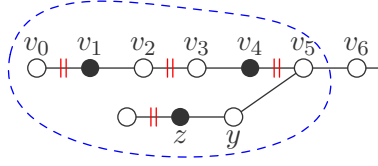


Figure 10: One possible choice of S

389 If y does not exist, then induct similarly with $S = \{v_0, \dots, v_5\}$.

390 This concludes the proof. QED

391 5 Trees where Matching Removal Has Little Impact

392 We considered earlier the case where the removal of a matching doubles or nearly
 393 doubles the domination number. We consider here the other extreme but only
 394 for trees. We show first that there is always a matching whose removal increases
 395 the domination number.

396 We will need the following fact, originally obtained by Bollobás and Cockayne:

397 **Lemma 2** [1] *The domination number of a graph without isolated vertices is at*
 398 *most its matching number.*

399 This yields the following:

400 **Theorem 6** *Removing any maximum matching M from a nontrivial tree T in-*
 401 *creases the domination number.*

402 **PROOF.** It is immediate that $T - M$ has $|M| + 1$ components. Thus $\gamma(T - M) \geq$
 403 $|M| + 1$, since we need at least one vertex in each component. By the above
 404 lemma, $|M| \geq \gamma(T)$. QED

405 As noted earlier, it is not true that every maximal matching removal increases
 406 the domination number.

407 We next consider the trees where removing any maximal matching only in-
 408 creases the domination number by at most 1. We have already seen that the star
 409 is an example. We define two families of trees:

- 410 • Let $\mathcal{S}$ denote the set of all trees that have radius at most two and at most
 411 one vertex of degree more than 2. We call the vertex of degree more than 2
 412 the “hub”.
- 413 • Let $\mathcal{T}$ denote the set of all trees obtained from a tree of diameter 3 by
 414 subdividing each edge once. (Note that all trees in $\mathcal{T}$ have odd order.) We
 415 call the starting tree the *base tree*.

416 Figure 11 gives a picture of a tree in $\mathcal{S}$ and a tree in $\mathcal{T}$.

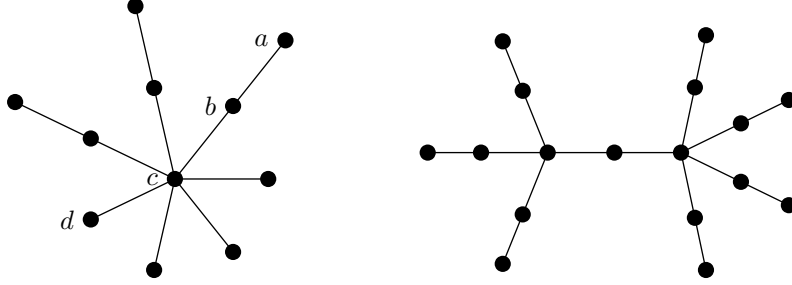


Figure 11: A tree in $\mathcal{S}$ and in $\mathcal{T}$

417 **Lemma 3** *If T is a tree in $\mathcal{S} \cup \mathcal{T}$, then the removal of a matching M from T*
 418 *either leaves the domination number unchanged or increases it by 1.*

419 **PROOF.** Assume $T \in \mathcal{S}$. Then $\gamma(T)$ equals the number of vertices of degree 2,
 420 plus one if the hub has a neighbor that is an end-vertex. Further, $\gamma(T - M)$ is
 421 at most the number of vertices of degree 2 plus one for the hub, plus one if M
 422 contains an edge joining the hub to an end-vertex neighbor.

423 Assume $T \in \mathcal{T}$ with order n . Then $\gamma(T) = (n-1)/2$ as, for example, the end-
 424 vertices and the central vertex form a dominating set while we need a different
 425 vertex to dominate each of them. On the other hand, $\gamma(T-M) \leq (n+1)/2$,
 426 as the original vertices of the base tree dominate T even after the removal of a
 427 matching. QED

428 It is well-known and trivial that adding an edge to a graph can reduce the
 429 domination number by at most 1.

430 **Theorem 7** *A tree T has the property that $\gamma(T-M) \leq \gamma(T)+1$ for all matchings*
 431 *M if and only if T is in $\mathcal{S} \cup \mathcal{T}$.*

432 **PROOF.** Let us say a matching is “bad” if its removal increases the domination
 433 number by more than 1, and a tree is “good” if it has no bad matching. By
 434 Lemma 3, it remains to show that if a tree T is good then it is in one of the two
 435 families. The proof is by induction on the order of T .

436 For the base case consider a tree of diameter at most 3. If such a tree has at
 437 most one vertex of degree more than 2, then it is in $\mathcal{S}$. If such a tree has two
 438 vertices of degree more than 2, then a maximum matching is bad. Hence we may
 439 assume T has diameter at least 4.

440 Consider a longest path P of good T with penultimate edge uv where all
 441 other neighbors of v are end-vertices. Say $T-uv$ consists of trees T_u and T_v . By
 442 Theorem 6, there exists a matching of T_u whose removal increases the domination
 443 number of T_u ; let M_u be any such matching (not necessarily maximum). Extend
 444 M_u to matching M of T by adding a leaf-edge incident with v . Then

$$\gamma(T-M) \geq \gamma(T_u-M_u) + 1 > \gamma(T_u) + 1 \geq \gamma(T).$$

445 In particular, $\gamma(T-M) = \gamma(T) + 1$ requires both that $\gamma(T) = \gamma(T_u) + 1$ and
 446 that $\gamma(T_u-M_u) = \gamma(T_u) + 1$. Since M_u was any matching such that $\gamma(T_u-M_u) >$
 447 $\gamma(T_u)$, this implies that T_u is good, and thus $T_u \in \mathcal{S} \cup \mathcal{T}$.

448 Let D_ℓ denote the set of support vertices of T , let M_ℓ be a matching consisting
 449 of a leaf-edge incident with each support vertex, and let U_ℓ be the tree obtained
 450 from T by removing each end-vertex incident to an edge of M_ℓ . We say that the

tree T is *leafy* if D_ℓ forms a dominating set. If so, $\gamma(T - M_\ell) = |M_\ell| + \gamma(U_\ell) = \gamma(T) + \gamma(U_\ell)$. So U_ℓ must have $\gamma(U_\ell) = 1$ and thus be a star if T is leafy.

Assume first that $T_u \in \mathcal{S}$. If T_u is a star, then T is leafy and $\gamma(T) = 2$. It is easily checked that the only way T can be good is that it is P_5 , which is in $\mathcal{S}$. So assume T_u is not a star. In T_u , let a be a vertex at distance 2 from the hub, c the hub, and b their common neighbor. Let d be an end-vertex neighbor of the hub, if it exists.

Up to symmetry there are four possibilities for u . (a) Assume $u = a$. Then $vabc$ is in U_ℓ , and so T is not leafy; thus the hub has no end-vertex neighbor. So if v has degree 2 in T the tree T is in $\mathcal{T}$. Otherwise it can be checked that any maximal matching containing the edge bc and a leaf-edge incident with v is bad. (b) Assume $u = b$. Then T is leafy. The only way U_ℓ can be a star is that both v and c have degree 2 in T , and c 's other neighbor is an end-vertex; so T is in $\mathcal{S}$. (c) Assume $u = c$. Then T is leafy. If v has degree 2 in T then T is in $\mathcal{S}$; otherwise U_ℓ is not a star. (d) Assume $u = d$. Then T is leafy, but U_ℓ is not a star, which contradicts the above.

Assume second that $T_u \in \mathcal{T}$. By the choice of P , u cannot be the central vertex of T_u . If u is an end-vertex of T_u , then it is easily checked that $\gamma(T) = \gamma(T_u)$, which contradicts the above. If u is a subdivision vertex in T_u , then one can check that there is maximum matching whose removal increases the domination number of T by 2. If u is one of the large-degree vertices, then we are in $\mathcal{T}$ if v has degree 2 in T , and there is a bad matching otherwise. QED

6 Further Thoughts

There are several directions to consider further. For example, can sharper bounds be obtained for other families of graphs? Even for trees, one could restrict the maximum degree and then ask about better bounds for the impact of matching removal. There is also the question of an algorithm for finding the matching in a tree whose removal increases the domination number the most, and the complexity of that task in general graphs.

References

- [1] B. Bollobás and E.J. Cockayne, Graph-theoretic parameters concerning domination, independence, and irredundance. *J. Graph Theory* 3 (1979), 241–249.
- [2] R.C. Brigham, P.Z. Chinn, and R.D. Dutton, Vertex domination-critical graphs. *Networks* 18 (1988), 173–179.
- [3] J.F. Fink, M.S. Jacobson, L.F. Kinch, and J. Roberts, The bondage number of a graph. *Discrete Math.* 86 (1990), 47–57.
- [4] G. Gunther, B. Hartnell, L.R. Markus, and D. Rall, Graphs with unique minimum dominating sets, *Congr. Numer.* 101 (1994), 55–63.
- [5] D.P Sumner, Critical concepts in domination. *Discrete Math.* 86 (1990), 33–46.
- [6] U. Teschner, New results about the bondage number of a graph. *Discrete Math.* 171 (1997), 249–259.